

Generator Set → Spectrum → Spectral Field

Generator set S

$\text{Spec}(A_S)$

Field $K_S = \mathbb{Q}(\{\lambda\})$

18 full
(18 moves)

Rational
 $\lambda = 1 - k/9$
5 distinct values

$\mathbb{Q}$

12 quarter
(12 moves)

Rational
 $\lambda = 1 - k/6$
6 distinct values

$\mathbb{Q}$

6 half-turn
(6 moves)

Rational
 $\lambda = 1 - k/3$
3 distinct values

$\mathbb{Q}$

n=8
(8 moves)

Irrational
 $\lambda \in \mathbb{Q}(\sqrt{5})$
7 distinct values

$\mathbb{Q}(\sqrt{5})$

n=16
(16 moves)

Irrational
 $\lambda \in \mathbb{Q}(\sqrt{5})$
8 distinct values

$\mathbb{Q}(\sqrt{5})$

random 9
(9 moves)

Mixed field
 $\mathbb{Q} \subset K_S$
14 distinct values

$\mathbb{Q}(\zeta_n)^+$



face-symmetric



symmetry-broken



non-symmetric

Face symmetry $\Rightarrow$ rational spectrum ($K_S = \mathbb{Q}$) | Symmetry breaking $\Rightarrow$ field extension